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Dr. med. Felix Busch
Chair of Medical Informatics
Institute for AI and Informatics in Medicine (AIIM)
TUM School of Medicine and Health
Ismaninger Str. 22, 81675 Munich

Re: Doctoral position, xMDT-HPB — explainable multimodal AI for hepato-pancreato-biliary oncology

Dear Dr. Busch,

I am applying for the doctoral position in the xMDT-HPB project. I want to do this PhD, and to do it inside a structured clinical-informatics group, for a specific reason: I have spent several years building explainable AI systems for cancer on my own, and I have reached the limit of what independent work can teach me about the part that matters most here — making those systems trustworthy to clinicians inside a real multidisciplinary tumour board, against real clinical data and real standards. That is the education a structured doctorate at AIIM offers and independent work cannot, and it is the reason I am applying rather than continuing alone.

The project's central word is the one I have worked on hardest. xMDT-HPB is built around AI that clinicians can *understand and trust*. Most work on this responds to the black-box problem with post-hoc explanation — SHAP values, attention maps, saliency — which documents a model's behaviour without making the model itself auditable. I have spent my independent research building diagnostic systems that are auditable in a stronger sense. The Syndrome framework performs MHC binding prediction (AUC 0.81, validated on 2,847 IEDB peptides), neoantigen identification, and cancer immune-escape scoring using three-channel discrete Fourier transforms on amino-acid physicochemical properties, with *zero* free parameters trained on patient data: every prediction is a geometric operation reproducible from the algorithm alone, independent of any training cohort. I do not present this as the method xMDT-HPB should adopt — clinical text, imaging, and laboratory trajectories are different problems with their own mature approaches. I present it as evidence that I understand, concretely and from the building side, the difference between a model that is *explained* and one that is *explainable*, which is exactly the distinction this project has to get right for clinicians to trust its output.

Hepato-pancreato-biliary, specifically. The pancreatic component of HPB is where my existing work already touches the project's clinical domain. Syndrome's neoantigen and immune-escape analysis was developed with KRAS-mutant pancreatic ductal adenocarcinoma as a worked case — the immune evasion that makes PDAC so difficult is one of the disease states the framework was built to characterise. I am not coming to HPB

cancer cold; I have read and modelled in this space, and I would bring that grounding while learning the radiological and clinical-narrative modalities that are new to me.

Multimodal and longitudinal data. The project integrates clinical text, multimodal imaging, and laboratory trajectories. Multimodal fusion and longitudinal inference are where my independent work already lives. I have built systems that combine physically distinct signal channels into a single inference, and a framework for monitoring *laboratory trajectories* as bounded, uncertainty-honest regions rather than point estimates — inferring a full state from partial, noisy, longitudinal observation with a provable convergence guarantee. That last problem — reconstructing a trustworthy trajectory from incomplete clinical measurement — is structurally the laboratory-data problem xMDT-HPB faces, and it is one I have worked on directly.

Open-source, reproducible, benchmarked — how I already work. The project commits to releasing everything as open-source software and reusable artifacts, with reproducible experiments and benchmarking pipelines. This is not an adjustment for me; it is how I build. My work is public on GitHub, my validation suites are deterministic and seeded so that results reproduce exactly from a stated seed, and every claim I make is checkable against an explicit reference rather than asserted. I would contribute to the project’s benchmarking and quality-assurance work from day one, because reproducible, independently-verifiable results are the standard I already hold myself to.

Why the structured path, and why now. I was previously a doctoral candidate in computational lipidomics at TUM, where I worked on machine learning for large clinical omics cohorts and distributed inference across federated sites, before funding ended and I left the programme. Since then I have worked independently and built a great deal — but the thing I most want now is precisely what this position provides and independent work does not: structured supervision, a real interdisciplinary team of clinicians and engineers, and clinical data and standards (MII, FHIR) I can only learn inside a group that works with them. I am genuinely prepared to familiarise myself with medical standards, data protection, and good scientific practice, and to do the careful, documented, team-aligned work the project describes. I would rather ground my work in a real clinical setting under supervision than continue to extend it alone, and this project is exactly that setting.

Background. I hold an MSc in Computational Biology (Universität Tübingen) and an MSc in Pharmaceutical Biotechnology (HAW Hamburg), with a BSc in Biochemistry and Cell Biology (Jacobs University Bremen). I conducted doctoral research in Computational Lipidomics at TUM (2019–2021). I work daily in Python and have hands-on experience with machine-learning frameworks and the full software-engineering discipline the role asks for: version control, documentation, testing, and reproducible pipelines. I have lived and worked in Germany since 2011 and hold permanent residence, so there is no work-authorisation or sponsorship question. I completed both of my master’s degrees — Pharmaceutical Biotechnology and Computational Biology — entirely in German, and taught master’s-level material in German at TUM. Two full German-language degrees are, I would suggest, more direct evidence of the academic and clinical German this role requires than any certificate: I have already done sustained scientific work in German at exactly this level. English is native.

I would welcome the chance to discuss how this background fits xMDT-HPB, and I can start as soon as the project requires.

Yours sincerely,

Kundai Farai Sachikonye